

Basic versus applied research: Julius Sachs (1832–1897) and the experimental physiology of plants

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The German biologist Julius Sachs was the first to introduce controlled, accurate, quantitative experimentation into the botanical sciences, and is regarded as the founder of modern plant physiology. His seminal monograph *Experimental-Physiologie der Pflanzen* (Experimental Physiology of Plants) was published 150 y ago (1865), when Sachs was employed as a lecturer at the Agricultural Academy in Poppelsdorf/Bonn (now part of the University). This book marks the beginning of a new era of basic and applied plant science. In this contribution, I summarize the achievements of Sachs and outline his lasting legacy. In addition, I show that Sachs was one of the first biologists who integrated bacteria, which he considered to be descendants of fungi, into the botanical sciences and discussed their interaction with land plants (degradation of wood etc.). This “plant-microbe-view” of green organisms was extended and elaborated by the laboratory botanist Wilhelm Pfeffer (1845–1920), so that the term “Sachs-Pfeffer-Principle of Experimental Plant Research” appears to be appropriate to characterize this novel way of performing scientific studies on green, photoautotrophic organisms (embryophytes, algae, cyanobacteria).

book *Die Organische Chemie in ihrer Anwendung auf Agricultur und Physiologie* (Organic Chemistry in its Application to Agriculture and Physiology),¹ Liebig proposed a novel theory of plant nutrition, arguing that the chemical elements of Nitrogen (N), Phosphorus (P) and Potassium (K) are key components to support vegetative growth and crop production. In addition, he reported that plants acquire the elements Carbon (C) and Hydrogen (H) from the atmosphere, and water (H₂O), plus dissolved mineral salts, from the soil. Unfortunately, Liebig’s conclusions, notably his version of the “theory of mineral nutrition of plants,” were largely based on older experiments performed by other investigators, or of speculative nature².

However, Liebig’s political agenda to popularize the image of chemistry and its use in agriculture was, at least in part, responsible for the establishment of German academic research stations aimed at increasing crop production during the Industrial Revolution.³

In 1861, the 29-year-old Privatdozent (Lecturer) Dr. Julius Sachs was appointed as a teacher/researcher to the *Landwirtschaftliche Akademie zu Poppelsdorf/Bonn* (Agricultural Academy of Poppelsdorf/Bonn), an Institution that later became part of the University (Fig. 1). In a short Curriculum Vitae that he had to submit to the German government in Berlin, Sachs summarized his life as follows: “Ferdinand Gustav Julius Sachs, Dr. philos., teacher of natural sciences at the Agricultural College of Poppelsdorf, born Oct. Two, 1832 in Breslau, evangelical Christian. Father: Christian Gottlieb Sachs, Engraver in Breslau, deceased; Mother: Theresa Sachs, geb. Hofbauer,

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Introduction

The chemist Justus Liebig (1803–1873) was, together with his older colleague Carl Sprengel (1787–1859), one of the pioneers of an applied area of plant-based research that was known in the 19th century as “Agriculturchemie” (agricultural chemistry). In his seminal 1840-

also deceased in Breslau. Until my 6th year, I lived in Breslau . . . 1845 I attended the Gymnasium Elisabethanum and earned, over the next 5 years, a 'praemium pro studio et virtute'. After the death of my parents (1848) I was, due to lack of any means, forced to leave the Gymnasium, and followed Professor Purkinje, who moved to Prague. In this city, I earned my Maturitaets-Examination (high school diploma) at the Clementinum, was then, for 3 years, a student of 'higher philosophy' at the University of Prague, and earned, after the successful passing of the required 4 examinations, my Ph.D. (1856). One year later, I obtained the *venia legendi* for Prague University and remained in this city as a Dozent (Lecturer) of Plant Physiology. 1859 I accepted a position in Tharandt, where I remained until the end of 1860. In the following year (1861), I was a teacher of Physiology at a school in Chemnitz, but gave up this position to come to Poppelsdorf.

On March 28, 1861, I was appointed as Lecturer at the Agricultural Academy at Poppelsdorf. Salary: 800 Thaler. No other job; since May 18, 1861 married to Johanna née Claudius from Prague. No children, without wealth and debt" (translated from the German text, ref.4).

During his 6-year-tenure in Poppelsdorf/Bonn, Sachs published 46 scientific papers and worked on his most influential book, the *Handbuch der Experimental-Physiologie der Pflanzen* (Experimental Physiology of Plants)⁵ (Fig. 1). This monograph inaugurated a new branch of experimental botany⁶ that will be detailed in the next section.

The Multi-Author Handbook that was Never Finished

In 1857, Julius Sachs (Fig. 2) contacted his colleague Wilhelm Hofmeister (1824–1877), the discoverer of the

homology of the life cycles in bryophytes, pteridophytes, and coniferous seed plants. A few years later (1861), Hofmeister became the Editor of a Four-Vol.-monograph entitled *Handbuch der Physiologischen Botanik* (Handbook of Physiological Botany).⁷ Unfortunately, the *Handbuch*, as envisioned by the Editor Hofmeister in 1866, was never published as scheduled, because one of the invited authors, Thilo Irmisch (1816–1879), did not submit the text assigned to him.⁸

In 1877, after Hofmeister's death, A. de Bary (Strassburg) and J. Sachs (Wuerzburg) announced, in the preface to Vol. III, the formal completion of this multi-author-monograph. The five books (Vols. I – IV) were arranged by de Bary and Sachs as follows:

Vol. I: W. Hofmeister (1867/1868) *Die Lehre von der Pflanzenzelle* (Plant Cell Biology) (A)

Allgemeine Morphologie der Gewächse (General Morphology of Plants) (B)

Vol. II: A. de Bary (1866) *Morphologie und Physiologie der Pilze, Flechten und Myxomyceten*

(Morphology and Physiology of Fungi, Lichens and Myxomycetes)

Vol. III: A. de Bary (1877) *Vergleichende Anatomie der Vegetationsorgane der Gefäßpflanzen*

(Comparative Anatomy of the Vegetation Organs of Cryptogams)

Vol. IV: J. Sachs (1865) *Experimental-Physiologie der Pflanzen*

(Experimental Physiology of Plants)

Vols. I to IV (i.e., 5 separate books) were published by the Verlag Wilhelm Engelmann in Leipzig.

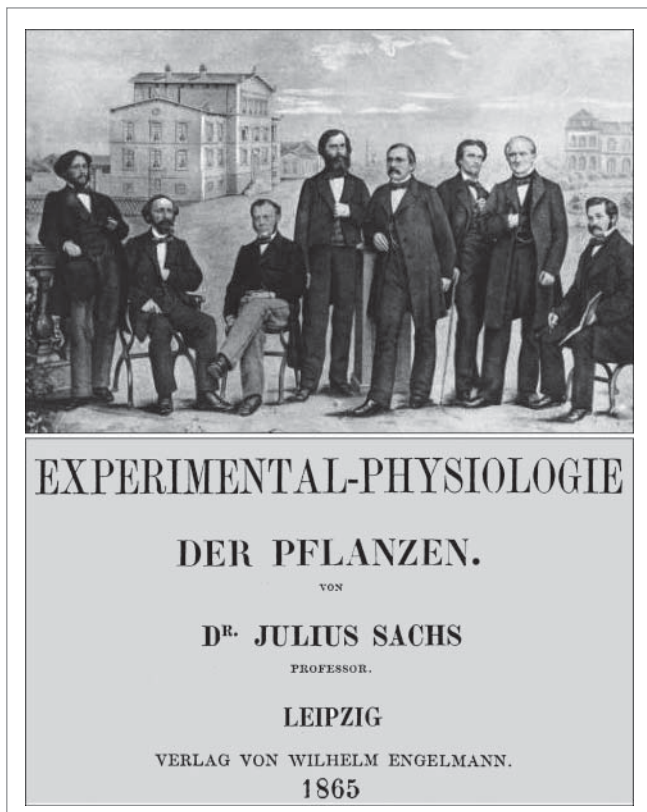


Figure 1. The 29-year-old Julius Sachs (1832–1897) (6th from the left) among some of his colleagues at the Agricultural Academy in Poppelsdorf/Bonn (i.e., the building in the background). During his 6-year-tenure, Sachs published, in addition to numerous journal articles, his textbook *Experimental Physiology of Plants* (1865) (adapted from ref.4).

This arrangement shows that 1. the two books of Hofmeister (1867/1868) were combined and issued as Vol. I; 2. the *Experimental-Physiologie* of Sachs (1865), which was published first, finally became Vol. IV, and 3. de Bary's monograph of 1877, with a concluding "Preface" signed by the author and Sachs on behalf of the deceased Hofmeister, represented Vol. III of this multi-author book.

The "Tables of Contents" of the "Handbook of Physiological Botany" (Vol. I to IV, 1865–1877)⁸ shows that 19th-century botanists studied all plant-like organisms that were not

unequivocally classified as animals: algae (inclusive of the blue greens, i.e., cyanobacteria), vascular cryptogams, bryophytes, angiosperms, lichens, fungi and plasmodial slime molds (myxomycetes). In 1872, the botanist Ferdinand Cohn (1828–1898) described microorganisms associated with plant material, and coined the name “Schizomyceten,” or “Spalt-Pilze” (Bakterien) for these tiny living beings.⁹ These prokaryotic microbes (that were discovered and described as “animalcules” in 1676 by Antonie van Leeuwenhoek, 1632–1723) were systematically investigated by botanists since ca. 1873.

This broad view of “the Plant Kingdom” is in contrast to the more specific opinion that Sachs (Fig. 2) expressed in his *Experimental-Physiologie*. Despite the fact that he referred, within the context of protoplasmic streaming, to de Bary’s work on the myxomycetes, the author largely focused on crop species, such as maize (“Turkish wheat”), bread wheat, sunflower, buckwheat, cucumber, broad bean etc. Accordingly, his favorite “green plants” were all characterized by oxygen-producing photosynthesis, a process he had studied over many years (for instance, light-induced accumulation of starch grains within the “chlorophyll bodies” of leaves; release of O₂-bubbles in irradiated aquatic plants that were maintained in CO₂-enriched water etc.).

A Novel Botanical Research Agenda

As mentioned above, the key publication of Sachs, his *Experimental-Physiologie der Pflanzen* (Experimental Physiology of Plants) was Vol. IV of an (unfinished) multi-author-monograph entitled “Physiological Botany.” In contrast to the book of Sachs (1865)⁵ (Fig. 1), which is still popular today, the other parts of Hofmeister’s series of monographs remained largely unknown. In his masterpiece, which was highly praised by Francis Darwin (1848–1925) and other well-known plant scientists, Sachs (1865) summarized all areas of plant research established at that time, and based his general conclusions mostly on his own

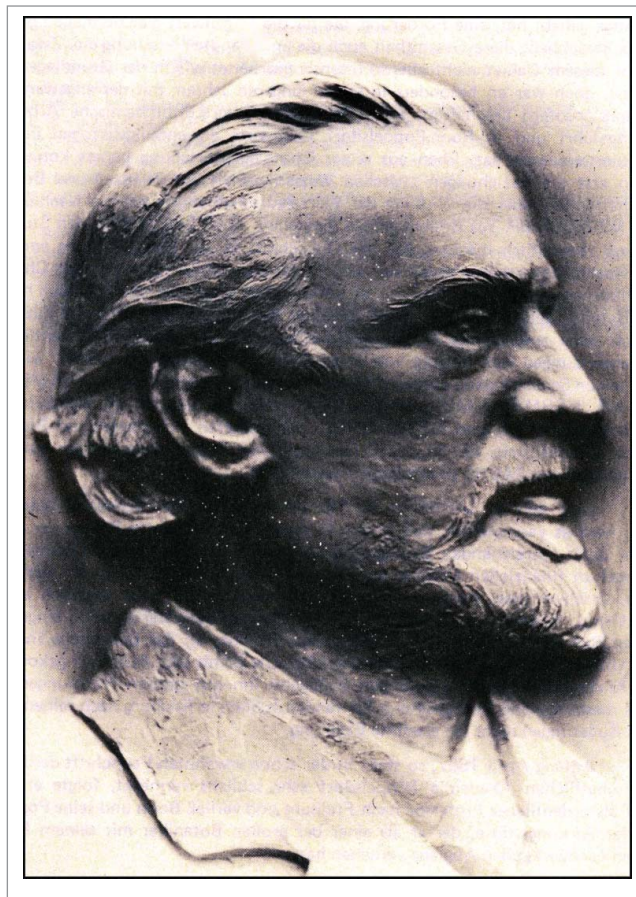


Figure 2. Julius Sachs (1832–1897), the founder of experimental plant physiology. Relief on the outside of the lecture hall, Institute of Agricultural Botany, University of Bonn, Germany (Artwork: A. Reusch) (adapted from ref. 7).

experimental studies. The author described, in chapters I to XIII, not only the effects of light, temperature, electricity, gravity, nutrients and atmospheric oxygen on physiological processes in plants, but also summarized the following topics: transformation of substances, translocation of organic material, molecular architecture of starch, and tissue tension in relation to organ growth. The book was published in November 1865 and was rapidly sold out. Translations into French (1868) and Russian (1867) made this publication well-known throughout many parts of Europe.¹⁰

Three key features characterize Sachs’ *Experimental Physiology* that distinguishes this monograph from all of its predecessors (for instance, the 2 books of Hermann Schacht [1814–1864] on the *Anatomy and Physiology of Plants*, 1856/1859, wherein “vegetation forces” etc. are discussed):⁶

1. In contrast to Schacht and others, Sachs (1865) did not mention “vital forces” etc.; he exclusively explained living processes in plants with reference to physical and chemical principles.
2. Sachs (1865) described novel methods and apparatuses for the experimental analysis of plant development and other physiological processes (water transport, transpiration, root pressure, germination, respiration, photosynthesis etc.). Moreover, he clearly pointed out that controlled, defined conditions (constant temperature etc.) are necessary to obtain reliable, reproducible, and hence meaningful results (Fig. 3). This was the major reason why the German biologist did not accept the “country-house-studies” of Charles Darwin and others¹¹ – Sachs had no trust in the experimental results obtained under variable environmental

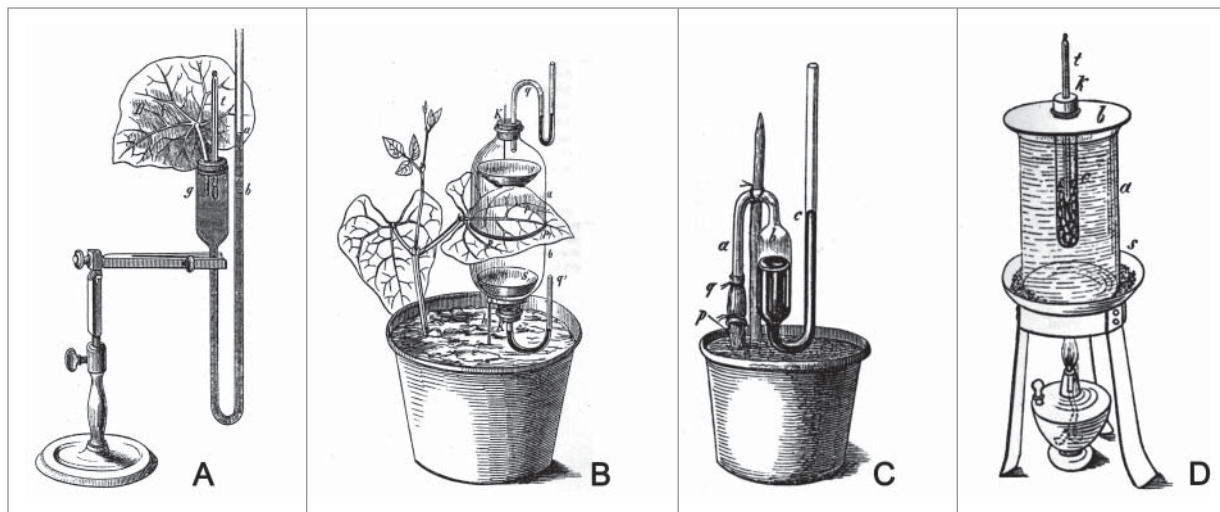


Figure 3. Custom-built apparatuses constructed and used by Julius Sachs during his tenure at the Agricultural Academy in Poppelsdorf/Bonn. Methods for the quantification of transpiration (A, B), root pressure (C), and the effect of temperature on seed germination (D) (adapted from ref.5).

conditions. In addition to physiological phenomena investigated on whole plants, Sachs also studied intracellular processes. For instance, in Chapter VII entitled *Molecularstruktur* he described and illustrated protoplasmic streaming in the hair cells of a squash (*Cucurbita pepo* L.) plant (Fig. 4). In this context, Sachs⁵ referred to the “Chlorophyllkörner” (chloroplasts) and compared these intracellular rotary movements with those observed in plasmodia of the myxomycetes.

3. Contrary to most other botanists of his time, Sachs⁵ made references to agriculture and practical applications of botanical studies. This may be due to the fact that Julius Sachs established his independent scientific career at institutions devoted to applied botany

and agricultural research (Fig. 1), rather than in classical Botanical Institutes at Universities.

For instance, Sachs analyzed the association of the root with soil particles in crop plants such as bread wheat (*Triticum aestivum* L.) (Fig. 5A), and discovered that the root hairs are largely responsible for the uptake of water and dissolved mineral salts (Fig. 5B). Together with Sachs’ well-known hydroculture-experiments, these studies established “root biology” as a new scientific discipline. As summarized by Hoexterma (1999)¹⁰ and others,^{11,12} most conclusions and theoretical concepts of Sachs concerning plant development, metabolism and behavior have been confirmed by subsequent investigators,^{13–16} with few exceptions (for instance, his

“imbibition theory of water transport”). Since the German botanist studied the physiology of crop plants with reference to agriculture, he established a new branch of the botanical sciences. This practical aspect of the work of Sachs is discussed in the next section.

Basic versus Applied Plant Research

In 1859, when Sachs was still a Lecturer at the University of Prague, the 27-year-old plant physiologist published a provocative thesis-paper in the journal *Der Chemische Ackersmann* (The Chemical Agriculturist) (Fig. 6). In this theoretical/philosophical contribution, Sachs (1859)¹⁷ argued that the science of plant physiology had been created by a few, curiosity-driven men (most of whom were poor), without financial aid from Government-supported Institutions. Moreover, he complained that “while everything in the world can be purchased; there are still many people who want to obtain knowledge, the most precious human product, free of charge” (ref.17).

Based on these premises, Sachs¹⁷ concluded that Agricultural Colleges, devoted to research dealing with the improvement of crop productivity, should employ not only chemists, but also plant physiologists. Moreover, he suggested that, in addition

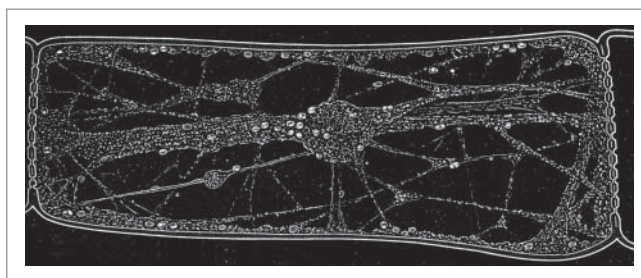


Figure 4. Drawing of a hair cell of a flower bud from squash (*Cucurbita pepo* L.), showing the phenomenon of protoplasmic streaming. The nucleus is in the center of the cell, and numerous chloroplasts are visible (adapted from ref.5).

to theoretical plant physiology, a second, applied branch should be established, which he labeled as “Agricultural Physiology.” According to Sachs, agricultural physiologists, who focus on crop plants, should get positions at research stations to supplement the work of chemists using “the physiological approach,” i.e., to elucidate open questions via “anatomical analyses and experiments” (Sachs 1859).¹⁷

It is obvious that Sachs’ concept of “agricultural (i.e., applied) plant physiology” was reminiscent to the ideas expressed in Liebig’s monograph of 1840, but the younger physiologist proposed a much more precise concept than the older chemist.

According to Sachs (1859)¹⁷, the cooperation of plant physiologists and chemists at Agricultural stations has the aim to improve and secure crop productivity. Under the headline “Tasks of the Agricultural physiologist,” he lists the following topics:

1. Seed germination (density of propagules, temperature, moisture, depth of the soil etc.).
2. Function of different plant organs during development (water- and nutrient uptake via the root system; leaves as assimilatory organs; senescence etc.).
3. Fruit development (role of plant nutrition; source and uptake of organic substances etc.) (ref.17).

Hence, the 27-year-old Julius Sachs defined plant physiology in 1859 as a basic and applied branch of the botanical sciences (Fig. 6). This novel view was elaborated and extended in his seminal *Handbuch* of 1865 (Fig. 1).

Recognition and Neglect of the Sachsian Research Agenda

Six decades after the publication of the *Handbuch* (Fig. 1), the *American Society of Plant Physiologists* (ASPP) (re-named in 2001 as the ASP Biologists, ASPB) was founded (1924), and in January 1926, Issue 1 of Vol. One of the new Journal *Plant Physiology* appeared in print. In the *Foreword*, the Editors explained the aims and scope of their new Periodical as

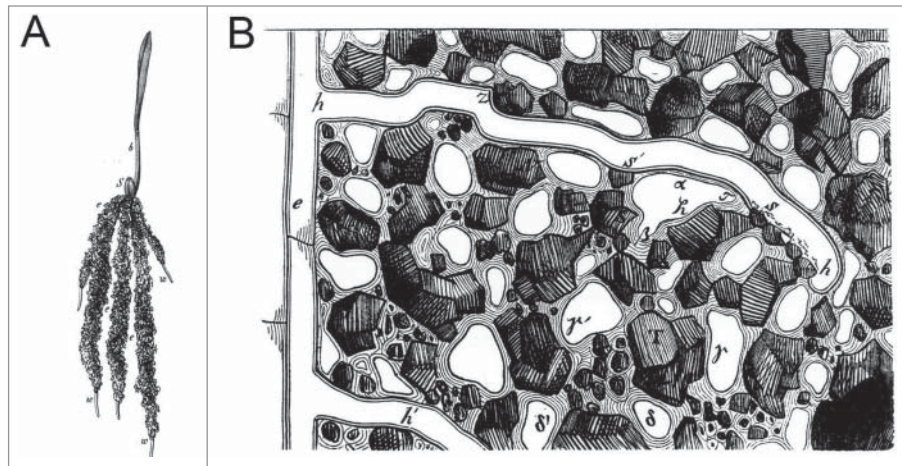


Figure 5. Drawing of a wheat (*Triticum aestivum* L.) seedling, with soil particles attached to the roots (A), and schematic rendering of the soil, penetrated by root hairs (B). Note that in this scheme the soil is composed of 3 phases: air-bubbles, capillary water and solid particles (adapted from ref.5).

follows: “With this issue, PLANT PHYSIOLOGY takes its place among the American journals published in the interests of botanical science. The Editors conceive their task as one of devoted service to the whole field of plant physiology;... Research in plant physiology must proceed in 2 general directions. It must continue to spread out into the practical fields of human service, such as agriculture, horticulture, agronomy, ecology, pathology, forestry, climatology, etc.; at the same time it must constantly delve more deeply

into the problems of developmental metabolism under the leadership of physiologists well trained in the methods of biophysics and biochemistry. Exploratory research, ... is of the utmost importance for the practical fields, for it yields us a broader knowledge of the methods of control of plant behavior and plant production. But exploratory research alone must lead only to empiricism, to rule of thumb methods of practice. Such exploratory research must be followed by an investigation of the fundamental causes of observed

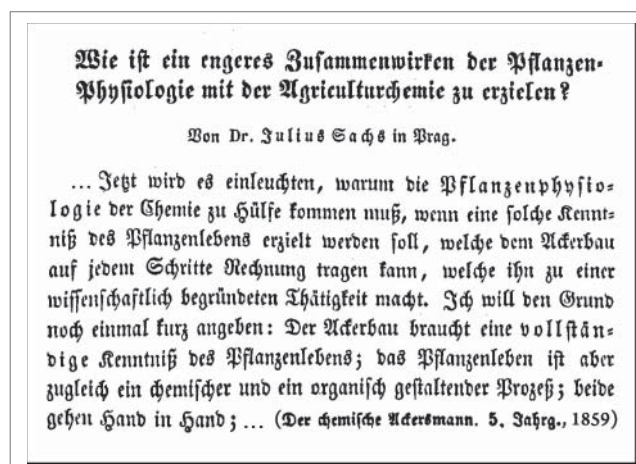


Figure 6. Headline (“How can a closer cooperation of Plant Physiology with Agricultural Chemistry be achieved?”) and key sentences of an important theoretical contribution of Julius Sachs published in 1859. The author argued that plant physiology and agricultural chemistry should join forces for the improvement of crop productivity.

behavior. . . . It is evident therefore that these 2 lines of investigation, practical and fundamental, must always go hand in hand. There can never be a logical separation of these 2 aspects of our science. Likewise, there can never be a logical separation of the pure physiologists from the practical physiologists. Our tasks are one, and we must learn to march together in their performance. . . . To this end it invites the support of plant physiologists of every denomination, ‘fundamentalists and modernists’, pure physiologists and applied physiologists. It has no other purpose, and no other desires than to be of service, and to promote cooperation in the

common tasks of advancing plant physiology as a pure and applied botanical science” (the Editors, 1926).¹⁸

It is obvious that in this anonymous *Editorial*, the vision of Sachs (1859)¹⁷ is expressed in words that are similar to those used by the German botanist decades earlier (Fig. 6). Accordingly, the life and scientific legacy of Julius Sachs was described in Vol. Four (1929) of the American journal *Plant Physiology*,¹⁹ a clear indication of the enormous international reputation of the German biologist.

In November 2014, the *Botanical Society of America* (BSA), established in 1893, published a Special Issue entitled

“Speaking of Food: Connecting Basic and Applied Plant Science.” In the Introduction, Gross et al. (2014)²⁰ argued that, since the “Food and Agricultural Organization of the United Nations predicts that food production must rise by 70% over the next 40 y to meet the demands of a growing population expected to reach 9 billion by the year 2050,” basic plant science is of great importance for agriculture. With reference to the work of the French chemist and bacteriologist Louis Pasteur (1822–1895), Gross et al. (2014)²⁰ distinguished between “Pure basic research (N. Bohr), Use-inspired basic research (L. Pasteur), and Pure applied research (T. Edison).”

However, the second concept, which has also been called “Pasteur’s quadrant” (Strokes 1997),²¹ is not new – it was proposed for the first time by Sachs (1859)¹⁷, and this principle of “use-inspired basic plant research” is described in detail in his *Experimental-Physiologie der Pflanzen* (Sachs 1865)⁵ (Fig. 1).

Louis Pasteur’s work was used by Strokes²¹ and Gross et al.²⁰ to illustrate the synthesis of basic and applied plant research, whereas the important contributions of Julius Sachs, the founder of modern plant physiology (Figs. 1, 2), were ignored. This may, in part, be due to the fact that Pasteur had studied bacteria, which were primarily viewed as pathogens of humans (“germ theory of disease”); hence, a medicinal aspect was associated with the research agenda of the French scientist that made his work much better known than that of the German botanist Julius Sachs.

Plants, Fungi and Bacteria

In his *Experimental-Physiologie der Pflanzen* of 1865, Sachs⁵ illustrated all basic living processes with reference to green algae and aquatic, as well as terrestrial plants (embryophytes). In addition, he sometimes referred to plasmodial slime molds (myxomycetes), and cited the work of his colleague A. de Bary in this context.²² Fungi are rarely mentioned (they were discussed in detail in de Bary’s monograph of 1866), and bacteria are absent in this 1865-book. This changed in

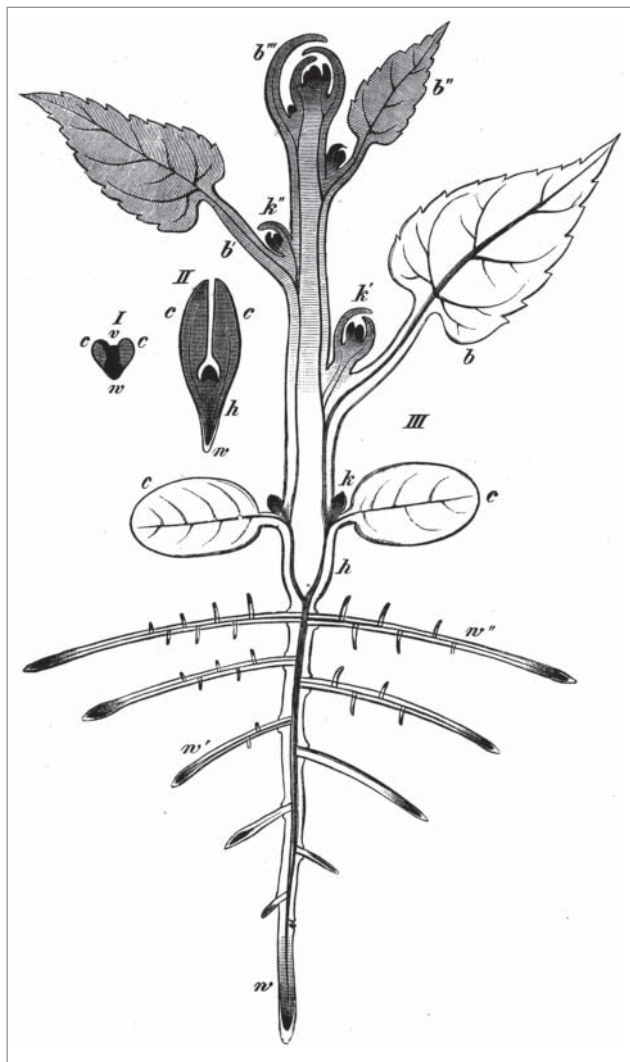


Figure 7. Scheme published by Julius Sachs in 1882, showing the basic body plan of an idealized dicotyledonous seed plant (III), with the development of the embryo (I, II). Note that the growing (meristematic) regions of this plant (buds, young leaves, root tips) are drawn in black/gray, whereas the mature parts of the organism are white (adapted from ref.23).

1872, after the plant scientist Ferdinand Cohn had introduced these microbes (“Schizomyceten,” also called “Fäulnis- or Spaltpilze”) into the botanical literature.

Interestingly, in the revised-, updated and extended version of his *Experimental-Physiologie*, the *Vorlesungen über Pflanzen-Physiologie* (Lectures on the Physiology of Plants), Sachs (1882)²³ discussed green plants (embryophytes), with reference to an idealized “model organism” (Fig. 7), as well as algae, lichens, fungi and bacteria (Fig. 8A and B). In his Lecture XXIV, Sachs (1882)²³ discussed the “Fäulnis- or Spaltpilze” (Bakterien) with reference to the work of F. Cohn. The author adopted the then-popular idea that bacteria are descendants of the hyphae of fungi (such as *Mucor*), and that the tiny microbes can also re-assemble to give rise to another fungus. These non-green “lower plants” (fungi and bacteria) are discussed with respect to the destruction of wood and other degenerative processes observed in rotten plant material, as well as putrefaction of fruits etc. Hence, Sachs (1882)²³ regarded both the fungi and bacteria as pathogenic microorganisms that destroy wood, fruits and other parts of healthy plants. Beneficial microbes (symbionts) are not mentioned by the author, although Sachs’ colleague de Bary had discussed symbiotic “plant-cyanobacteria” (*Azolla/Anabaena*) interactions in several of his publications.²²

Interestingly, Sachs (1882)²³ described in detail the thallus of liverworts, such as that of the widely distributed species *Marchantia polymorpha* (Fig. 9A and B). In this context, the botanist referred to the gametophyte (i.e., the green plant body) as representing the dominant phase of the life cycle in this “primitive land plant” and provided insights into the cellular structure of this organism that is characterized by dichotomously branched thalli (with rhizoids attaching them to the soil) that exhibit apical growth.

Ten years ago, it was discovered that the vegetative growth of pieces of thalli (or isolated vegetative propagules, the gemmae) of liverworts, such as *M. polymorpha*, is promoted by naturally-occurring, plant-associated methylobacteria.²⁴ These α -proteobacteria, which can be isolated/cultivated on agar-plates (Fig. 10A), consume

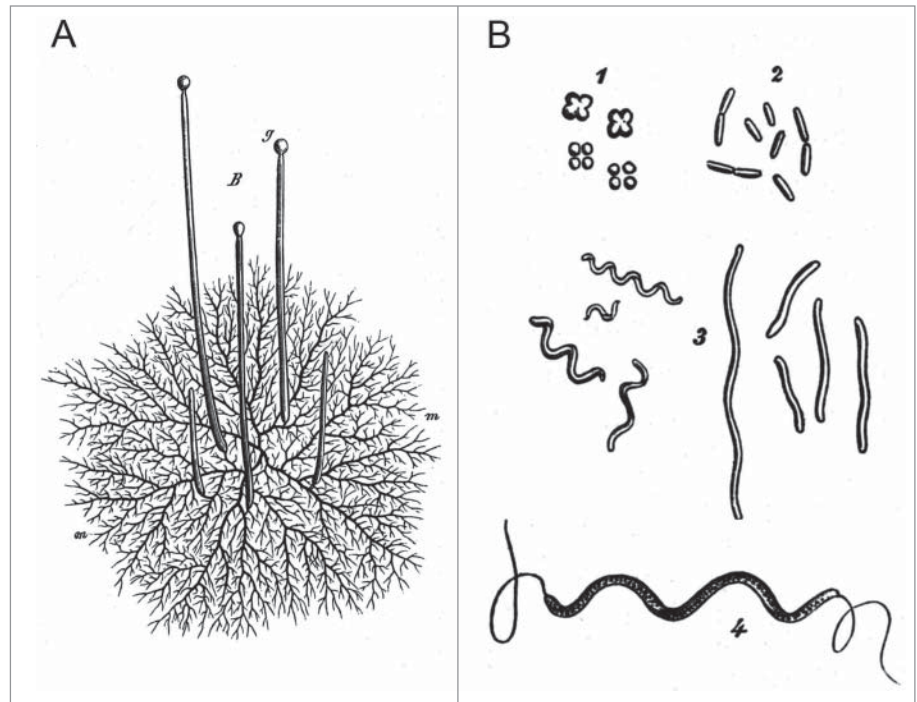


Figure 8. Drawings of the fungus *Phycomyces nitens*, with a network of hyphae (mycelium) and sporangia (A). The groups of bacteria 1 to 4 were described by Julius Sachs in 1882 as Schizomycetes (Spaltpilze) (B). Different morphotypes of bacteria are depicted schematically, with reference to the work of F. Cohn (adapted from ref.23).

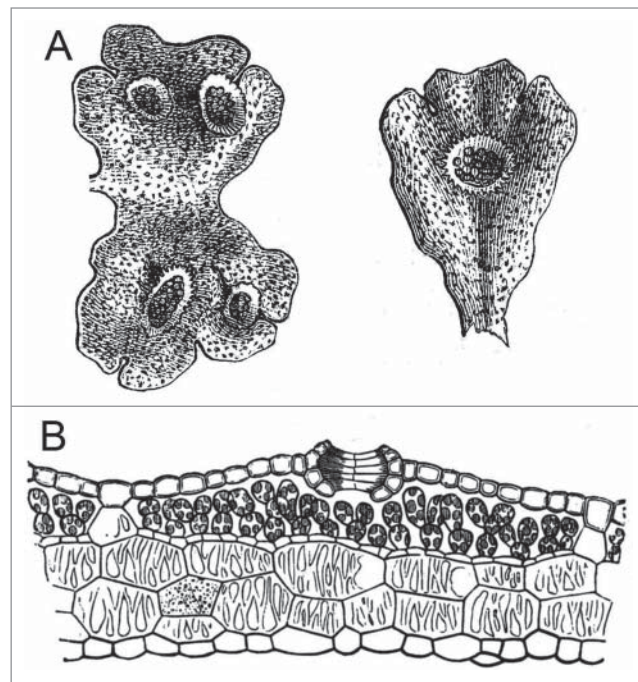


Figure 9. Julius Sachs frequently referred to “lower plants” (bryophytes etc.) to describe basic principles of growth and reproduction. Morphology of part of the thallus of a mature liverwort (*Marchantia polymorpha*) (gametophyte), without stalked sporophytes (archegonia). Shallow cups with disk-shaped gemmae (vegetative propagules) are visible (A). The cross-section through the thallus shows a respiratory pore, the upper/lower epidermis, and mesophyll tissues (B) (adapted from ref.23).

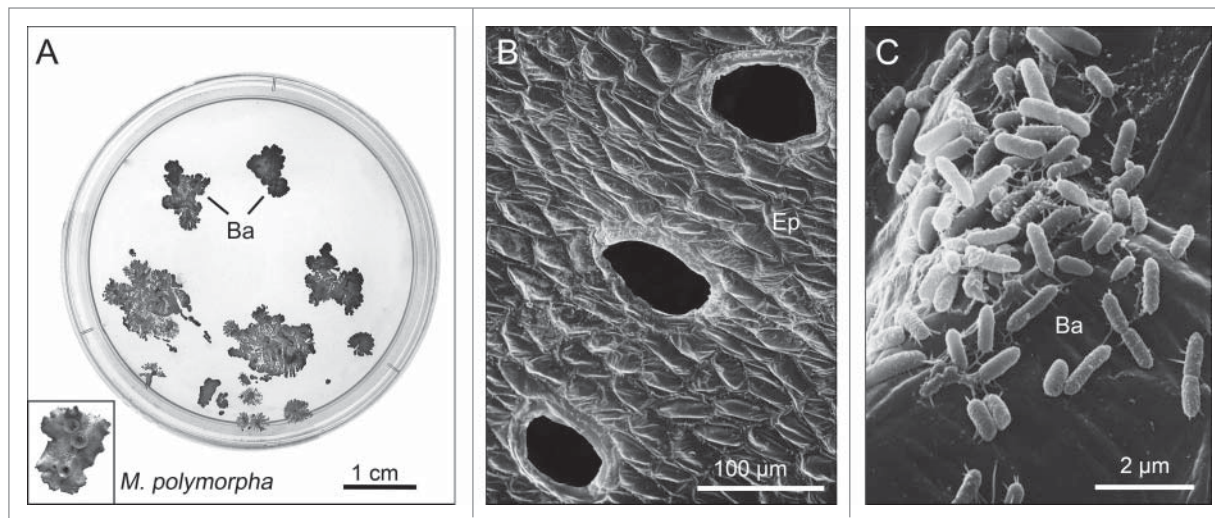


Figure 10. Photograph of a piece of the liverwort *Marchantia polymorpha* and agar plate-impressions, showing colonies of pink-pigmented methylobacteria isolated from the upper surface of the thallus (A). Scanning electron micrographs of the upper side of the thallus of *M. polymorpha*, with 3 respiratory pores (B). On the surface of the epidermal cells, and the margin of the pores, bacteria are attached. At 50-fold larger magnification, numerous epiphytic microbes (*Methylobacterium mesophilicum*) become visible. These bacteria live in a super-cellular biofilm, in which individual prokaryotes are attached to each other, and to the epidermal cells of the plant, via extracellular polymeric substances (thread-like structures) (C). Ba = bacteria, Ep = epidermis (unpublished results).

methanol released by the growing plant cells, and stimulate elongation growth of their host organism via the biosynthesis/secretion of phytohormones (auxins, cytokinins).²⁵ Figure 10B shows the upper surface of a thallus that grows rapidly in the presence of methylobacteria. At larger magnification, numerous epiphytic microbes can be detected that are attached to the epidermal cells of the liverwort via extra-cellular polymeric substances. Moreover, it is obvious that these plant-associated methylobacteria live in a social community, referred to as a “bacterial biofilm” (Fig. 10C). Based on these and other findings, these microbes have been classified as co-evolved phytosymbionts.^{24–28}

Julius Sachs’ younger colleague, the German botanist Wilhelm Pfeffer (1845–1920), elaborated and extended the Sachsonian principle of controlled experimentation. Moreover, Pfeffer fully incorporated bacteria as pathogens, commensals, or cooperation partners of green plants into botanical research. For instance, in his textbook *Pflanzenphysiologie* (Vol. I, 1897/Vol. II, 1894),²⁹ Pfeffer discussed Theodor Engelmann’s (1843–1909) “bacteria-experiments” of 1882 for the elucidation of the action spectrum of photosynthesis in filamentous green algae,

such as *Oedogonium*. This outstanding discovery was not accepted by Sachs, who responded with polemical remarks directed against Engelmann (and others) when these findings were published.⁶

In addition, Pfeffer (1897/1904)²⁹ described the symbiotic relationship between leguminous plants (*Lupinus luteus* etc.) and soil-borne bacteria of the genus *Rhizobium* with respect to nitrogen fixation. Hence, Wilhelm Pfeffer was the most innovative and creative successor of Julius Sachs, so that it is appropriate to introduce the term “Sachs-Pfeffer-Principle of Experimental Plant Research.” However, in contrast to Sachs, who was a gifted author, Pfeffer’s work is difficult to read, due to his formal style. Nevertheless, Pfeffer should be regarded as the co-founder of experimental plant biology,³⁰ notably since he fully incorporated the “microbiological aspects” into this emerging scientific discipline.

Conclusions

In his *Review* of E. G. Pringsheim’s monograph on Julius Sachs, the botanist and psychologist Arthur Tansley (1871–1955) summarized the rare combination of characters likely responsible for Sachs’ genius as

follows: 1. He was a great investigator with the urge to go direct to nature and find out for himself; 2. he was a gifted experimentalist, with the ability to devise and construct his own apparatuses; 3. he had the true philosophical mind which always and everywhere seeks the general significance of particular phenomena, and 4. he had the spirit of the creative artist who aims at constructing an artistic whole from the materials with which he works.³¹ Hence, the specific quality of the publications authored by Julius Sachs may rest on these 4 specific characters of the great scientist (and philosopher/artist).

In 1866, Sachs left the Agricultural College in Poppelsdorf/Bonn (Fig. 1) to become, as successor of Anton de Bary, Professor of botany at the University of Freiburg i. Br., where he wrote his influential *Lehrbuch der Botanik* (Textbook of Botany) (1868; 4. Ed. 1874).³² After only 3 semesters, he accepted the chair of botany at the University of Wuerzburg, where he stayed until his death on May 29, 1897. In Wuerzburg, Sachs published a book on the *Geschichte der Botanik* (History of Botany) (1875),³³ and his famous *Vorlesungen über Pflanzen-Physiologie* (Lectures on the Physiology of Plants) (1882; 2. Ed. 1887).²³

Julius Sachs was the mentor of numerous well-known botanists, such as Francis Darwin and Wilhelm Pfeffer; he also exerted a strong influence on Arthur Tansley, who wrote, with reference to the English version (1887) of Sachs' *Vorlesungen*, that "It was this translation, read when it appeared in 1887, that first attracted interest of the present reviewer, as a boy, to scientific botany." (ref.31). As a result of his outstanding achievements, Sachs received many honors and awards during the second half of his professional career. However, despite this world-wide recognition, he remained a dedicated, creative, hard-working scientist and evolutionist thorough his life.³⁴⁻³⁶

Finally, it should be stressed that his *Vorlesungen*, published in 1882, may be interpreted as the "second, updated/extended edition" of Sachs' *Experimental-Physiologie* of 1865. In this elegant, well-illustrated book, wherein he published his most famous drawing, a general scheme of a dicotyledonous plant (Fig. 7), Sachs also discussed bacteria with reference to the living processes of angiosperms (Fig. 8). However, it was Wilhelm Pfeffer, who fully incorporated the "bacterial world" into 19th-century botanical sciences. Therefore, Pfeffer should be honored as the co-founder of experimental plant physiology, which can be defined in 2015 as "systems biology of photoautotrophic organisms (embryophytes, algae, and cyanobacteria)."

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